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Microbiota Disruption Induced by Early Use of Broad-Spectrum Antibiotics Is an Independent Risk Factor of Outcome after Allogeneic Stem Cell Transplantation

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A B S T R A C T

In allogeneic stem cell transplantation (ASCT), systemic broad-spectrum antibiotics are frequently used for treatment of infectious complications, but their effect on microbiota composition is still poorly understood. This retrospective analysis of 621 patients who underwent ASCT at the University Medical Center of Regensburg and Memorial Sloan Kettering Cancer Center in New York assessed the impact of timing of peritransplant antibiotic treatment on intestinal microbiota composition as well as transplant-related mortality (TRM) and overall survival. Early exposure to antibiotics was associated with lower urinary 3-indoxyl sulfate levels ($P < .001$) and a decrease in fecal abundance of commensal *Clostridiales* ($P = .03$) compared with late antibiotic treatment, which was particularly significant ($P = .005$) for *Clostridium* cluster XIVa in the Regensburg group. Earlier antibiotic treatment before ASCT was further associated with a higher TRM (34%, 79/236) compared with post-ASCT (21%, 62/297, $P = .001$) or no antibiotics (7%, 6/88, $P < .001$). Timing of antibiotic treatment was the dominant independent risk factor for TRM (HR, 2.0; $P \leq .001$) in multivariate analysis besides increase age (HR, 2.15; $P = .004$), reduced Karnofsky performance status (HR, 1.47; $P = .03$), and female donor–male recipient sex combination (HR, 1.56; $P = .02$) A competing risk analysis revealed the independent effect of early initiation of antibiotics on graft-versus-host disease–related TRM ($P = .004$) in contrast to infection-related TRM and relapse (not significant). The poor outcome associated with early administration of antibiotic therapy that is active against commensal organisms, and specifically the possibly protective *Clostridiales*, calls for the use of *Clostridiales*-sparing antibiotics and rapid restoration of microbiota diversity after cessation of antibiotic treatment.

INTRODUCTION

Allogeneic stem cell transplantation (ASCT) is a curative treatment option for a variety of hematopoietic malignancies and other severe hematologic and genetic diseases [1]. Its success, however, is still limited by a significant risk of life-threatening complications, including acute graft-versus-host disease (GVHD) and infection. Because neutropenia and mucosal injury after myeloablative conditioning regimens often lead to neutropenic infections in ASCT recipients [2], most patients require treatment with broad-spectrum antibiotics. However, there is increasing evidence that use of broad-spectrum antibiotics may exert a detrimental impact on intestinal microbiota composition and, subsequently, the outcome of ASCT [3]. Studies have demonstrated that loss of

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intestinal microbiome diversity, in particular commensal *Clostridiales*, appears to contribute not only to the pathogenesis of acute gastrointestinal GVHD but also to increased transplant-related mortality (TRM) [4,5]. Similarly, urinary concentration of 3-indoxyl sulfate (3-IS), a tryptophan metabolite of commensal colonic bacteria, has been identified as an indirect marker of a balanced microbiota and predicts outcome at the time of ASCT [6]. In the present study we investigated the association of the timing of antibiotic treatment with intestinal microbiota composition and clinical outcome in a cohort of 621 patients undergoing ASCT in 2 centers with different practices of antibiotic prophylaxis.

METHODS

Patients

A total of 621 adult patients undergoing ASCT in Regensburg, Germany (n = 380) and New York, New York (n = 241) were included in our retrospective analysis. Inclusion criteria were hemato-oncologic disease requiring ASCT with an age above 18 years and receipt of non–T cell–depleted grafts. The Regensburg cohort consisted of 380 consecutive ASCT recipients enrolled between September 2008 and June 2015, whereas the New York cohort comprised 241 patients who underwent ASCT between October 2009 and May 2015 at the Memorial Sloan Kettering Cancer Center in New York. Further, with approval by the Ethics Committee of the University Medical Center of Regensburg and after receipt of written informed consent, urinary and stool specimens were collected in a total of 130 and 26 patients, respectively, at a minimum of 6 different time points between admission and day 28 after ASCT: before admission, at least once between days −2 to +2, +2 and +10, +11 to +17, +18 to +24, and +25 to +30. In addition, stool samples from 5 healthy stem cell donors served as controls. Similarly, for the New York group stool specimens from 146 ASCT recipients were obtained within 4 days of day 12 after ASCT. All specimens were stored at −80°C until analysis.

All patients received prophylactic antibiotics from the start of conditioning until engraftment, but the type of prophylactic antibiotics differed between the 2 cohorts. In Regensburg, ciprofloxacin 500 mg twice daily and metronidazole 400 mg three times daily were administered orally to 189 patients (49.7%) until March 2012, before prophylaxis was switched to oral rifaximin 200 mg twice daily in the remaining 191 patients (50.3%) to control the emergence of vancomycin-resistant enterococci. In the New York group, all 241 patients received prophylactic intravenous ciprofloxacin 400 mg twice daily, and those receiving myeloablative preparative regimens were additionally given intravenous vancomycin at a starting dose of 1 g twice daily.

Although there were differences in prophylactic antibiotic regimens between the New York and Regensburg cohorts, treatment of neutropenic fever/infections with additional antibiotics was comparable between the 2 cohorts. In the Regensburg cohort 350 of 380 patients (92%) received additional antibiotic treatment beyond prophylactic regimens. Piperacillin/tazobactam at a thrice-daily dose of 4.0/.5 g was used as empiric first-line therapy in 275 of 350 patients (78%), whereas meropenem 1.0 g thrice daily and vancomycin 1.0 g twice daily served as second-line therapy in 55 of 350 patients (16%). In cases of penicillin intolerance, 20 patients (6%) received alternative first-line treatment with vancomycin and/or ceftazidime (in 13 patients after ciprofloxacin/metronidazole prophylaxis; in 7 patients after rifaximin prophylaxis). In the New York cohort, piperacillin/tazobactam and imipenem/cilastatin were administered 4 times a day at doses of 4.0/.5 g and 500 mg, respectively, as first- and second-line therapy. Overall, 183 of 241 patients (76%) in New York required antibiotics to treat neutropenic fever/infections, among whom 103 (56%) received first-line therapy only and 80 (44%) required second-line therapy. Fifty-eight of 241 ASCT recipients (24%) required no therapeutic antibiotics. The clinical criteria for initiation of first- and second-line antibiotics were comparable with the Regensburg group. At both centers patients received first- and second-line treatment according to international guidelines for treatment of neutropenic fever/infections [7,8].

All patients were classified into 3 groups according to the timing of additional antibiotic initiation: (1) early exposure to antibiotics between days −7 and 0 (early AB group, n = 236 [38%]), (2) exposure to antibiotics on day 0 or thereafter (late AB group, n = 297 [48%]), and (3) no systemic antibiotic treatment during the course of ASCT beyond prophylactic regimens (no AB group, n = 88 [14%]) (Table 1). Because acute leukemia patients usually have a history of several courses of cytotoxic treatment and repeated infections treated with broad-spectrum antibiotics, we additionally defined 2 subgroups and divided patients into an acute leukemia (n = 296) versus nonacute leukemia (n = 325) group. Patient characteristics in the 3 antibiotic subgroups are shown in Table 2.

Table 1
Beginning of Antibiotic Treatment at the 2 Centers

	Total Cohort (N = 621)	Regensburg (n = 380)	New York (n = 241)
No AB group	14% (n = 88)	8% (n = 30)	24% (n = 58)
Early AB group	38% (n = 236)	50% (n = 190)	19% (n = 46)
Late AB group	48% (n = 297)	42% (n = 160)	57% (n = 137)

Analysis of Urinary 3-IS

Urinary 3-IS and creatinine levels were determined by reverse-phase liquid chromatography-electrospray ionization-tandem mass spectrometry as previously described [6].

Quantification of Clostridium *Cluster XIVa 16S rRNA Gene Copies by Reverse Transcriptase PCR*

Using *Clostridium* cluster XIVa group-specific primers and SYBR Green I Master (Roche, Basel, Switzerland) qPCR reagents, 16S rRNA gene copy numbers of *Clostridium* cluster XIVa species were determined in fecal DNA preparations by real-time quantitative PCR on a LightCycler 480 II instrument (Roche). Full-length 16S rDNA amplicons of *Clostridium* cluster XIVa bacteria cloned into the pGEM T-Easy vector served as quantification standards (Invitrogen, Carlsbad, CA).

Clostridial Abundance Measurement

At Memorial Sloan Kettering Cancer Center, stool specimens collected within 4 days of day 12 after ASCT were analyzed for clostridial abundance, as described previously [9]. Briefly, stool specimens were subjected to mechanical disruption (bead-beating), and DNA was extracted with phenol-chloroform [10]. DNA samples were analyzed by the Illumina MiSeq platform (Illumina, San Diego, CA) to sequence the V4-V5 region of the 16S rRNA gene. Sequence data were compiled and processed using mothur version 1.34 [11], screened and filtered for quality [12], and classified to the species level [13] using a modification of the Greengenes reference database [14].

Bioinformatics and Data Analysis

Normally and non-normally distributed continuous data are presented as mean (± standard deviation [STD]) or median (range), respectively. Accordingly, group comparisons were performed by 2-sided *t*, Mann-Whitney U. or Kruskal-Wallis tests. Absolute and relative frequencies were given for categorical data and compared between study groups by chi-square or Fisher's exact tests. All hypotheses were tested in an explorative manner on a 2-sided 5% significance level. Factors associated with early use of systemic antibiotics were assessed using logistic regression analysis. Kaplan-Meier analysis was performed to assess survival and nonrelapse mortality, and Cox regression was used for multivariate assessment of risk factors. Competing-risk analysis [15] for GVHD-related TRM, infectious TRM, and relapse was performed using software package R 3.2.2 (The R Foundation of Statistical Computing, Vienna, Austria). Otherwise, IBM SPSS Statistics 22 (SPSS Inc., Chicago, IL) was used for analysis.

RESULTS

Factors Associated with Early Antibiotic Treatment in ASCT Patients

In the study cohort, multivariate analysis identified several factors associated with higher likelihood of early antibiotic exposure before day 0: advanced stage of underlying disease (hazard ratio [HR], 2.1; 95% confidence interval [CI], 1.7 to 2.7; $P < .001$), matched unrelated donor (HR, 2.2; 95% CI, 1.5 to 3.4; $P < .001$), and interval from first diagnosis to ASCT < 12 months (HR, 2.6; 95% CI, 1.8 to 3.8; $P < .001$). Patient age and donor–recipient sex ratio did not affect timing of antibiotic treatment. The interval from diagnosis to ASCT was shorter in patients suffering from acute leukemia with a median of 5 months (range, 1 to 206) than in nonacute leukemia patients requiring ASCT with a median of 18 months (range, 2 to 156; $P < .001$).

Timing of Antibiotics Affects Intestinal Microbiota Disruption Early after ASCT

Several lines of evidence suggest an impact of early antibiotics on loss of commensal bacteria and therefore raise

Table 2
Summary of Patient Characteristics in the 3 AB Subgroups

	Early AB group (n = 236)	Late AB group (n = 297)	No AB group (n = 88)
Type of underlying disease			
Acute leukemia	57.6% (n = 136)	43.8% (n = 130)	34.1% (n = 30)
No acute leukemia	42.4% (n = 100)	56.2% (n = 167)	65.9% (n = 58)
Age of the patient, yr			
<20	1.3% (n = 3)	.7% (n = 2)	0% (n = 0)
20-40	16.1% (n = 38)	19.5% (n = 58)	13.6% (n = 12)
>40	82.6% (n = 195)	79.8% (n = 237)	86.4% (n = 76)
Donor type			
HLA-identical sibling donor	19.5% (n = 46)	25.3% (n = 75)	53.4% (n = 47)
Unrelated donor	80.5% (n = 190)	74.7% (n = 222)	46.6% (n = 41)
Donor–recipient sex combination			
Donor female, male recipient	16.9% (n = 40)	22.2% (n = 66)	21.6% (n = 19)
All other	83.1% (n = 196)	77.8% (n = 231)	78.4% (n = 69)
Disease stage			
Early	20.8% (n = 49)	33.6% (n = 100)	43.2% (n = 38)
Intermediate	32.2% (n = 76)	36.4% (n = 108)	36.4% (n = 32)
Late	47.0% (n = 111)	30.0% (n = 89)	20.4% (n = 18)
Time interval from diagnosis to transplant, mo			
<12	64.4% (n = 152)	55.2% (n = 164)	52.3% (n = 46)
≥12	35.6% (n = 84)	44.8% (n = 133)	47.7% (n = 42)
Karnofsky performance status			
≥90%	62.7% (n = 148)	66.3% (n = 197)	62.5% (n = 55)
<90%	37.3% (n = 88)	33.7% (n = 100)	37.5% (n = 33)

the hypothesis that antibiotic-induced intestinal dysbiosis was involved in the poor outcome of patients receiving early antibiotic treatment. Because urinary levels of 3-IS directly correlate with the abundance of *Clostridiales* [7], we measured the levels of this microbial biomarker in patients with assessable samples from the Regensburg cohort. At baseline, no differences in 3-IS levels were observed before ASCT between the 3 antibiotic subgroups (early AB group: 30.3, STD 46.1 μmol/mmol creatinine; late AB group: 24.9, STD 39.4 μmol/mmol creatinine; no AB group: 19.0, STD 8.0 μmol/mmol creatinine; not significant). In contrast, significantly lower mean urinary concentrations of 3-IS between days 0 and 28 after ASCT were found in the early AB group (n = 62, mean 5.5, STD 13.6 μmol/mmol creatinine) than in late AB (n = 57, mean 9.9, STD 8.2 μmol/mmol creatinine; $P < .001$) and no AB groups (n = 11, mean 11.1, STD 10.5 μmol/mmol creatinine; $P = .02$), respectively, in the Regensburg cohort. The trend of 3-IS levels in the different subgroups from pretransplant to day 28 after ASCT is shown in Figure 1.

In the New York cohort, clostridial abundance was analyzed by 16S amplicon sequencing and was found to be higher in the late AB group (2.3%) than in the early AB group (.74%,

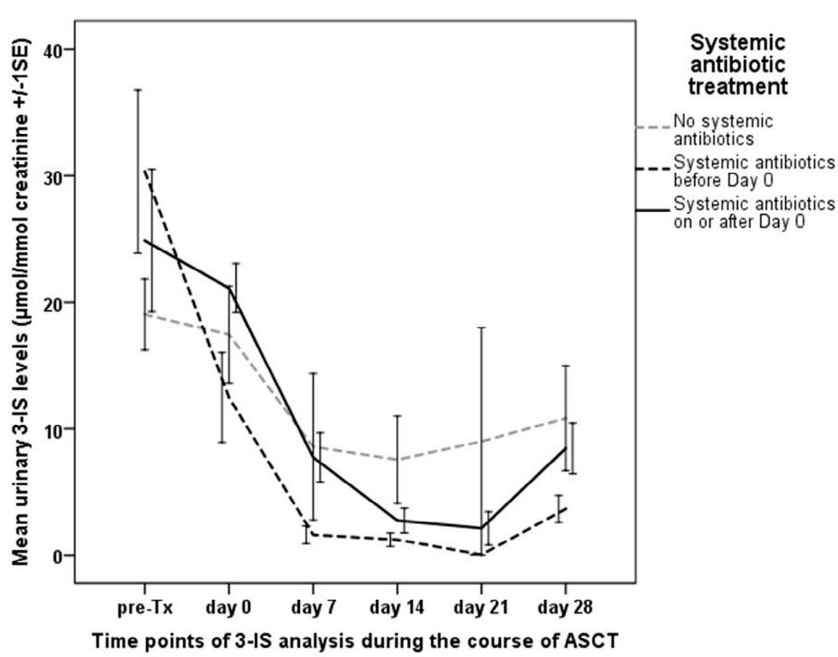


Figure 1. Impact of antibiotic treatment on the course of urinary 3-IS levels during the first 28 days after ASCT. Mean urinary 3-IS levels between days 0 and 28 after ASCT indicate signficant long-term suppression of commensal *Clostridiales* by early antibiotic treatment.

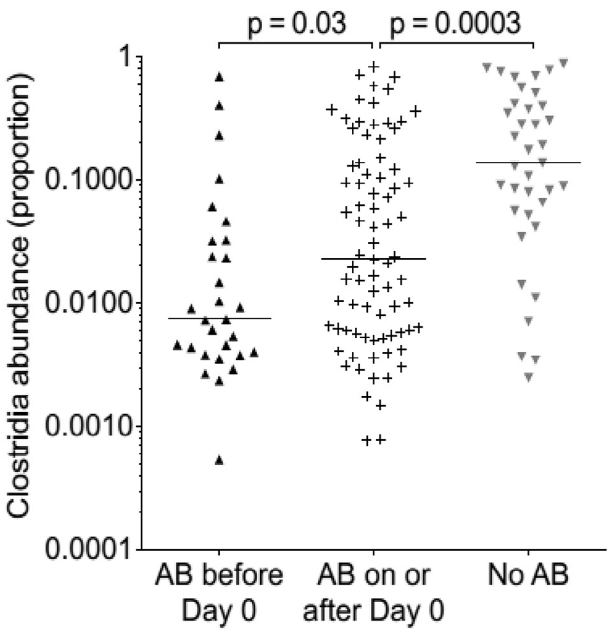


Figure 2. Comparison of Clostridial abundance in stool specimens in relation to timing of antibiotics. Clostridial abundance was higher in the late or no AB group compared with the early AB group.

$P = .03$). Patients that had received no therapeutic antibiotics showed the highest clostridial abundance (14%, $P = .003$, Figure 2).

In a small subgroup of Regensburg patients who received rifaximin prophylaxis, we performed direct quantification of *Clostridium* cluster XIVa 16S rRNA gene copies by reverse transcriptase PCR in stool specimens collected from the early (n = 13) and late (n = 13) AB groups. Between days 0 and 7 after ASCT, significantly lower copy numbers were found in the early AB group (2.4×10^4) compared with the late AB group (2.6×10^8, $P = .005$), indicating directly that early antibiotic treatment markedly reduced this subset of commensal intestinal bacteria. A direct comparison with healthy donors (n = 5) showed significantly higher copy numbers with 3.4×10^9 copy numbers.

Early Start of Systemic Antibiotics Is Associated with Poor Outcome after ASCT

Regarding outcome and early systemic antibiotic treatment, we found a significantly (log rank = .001) increased TRM in the early AB (34%, 79/236) compared with the late AB group (21%, 62/297). The lowest rate of TRM (7%,= 6/88) was found in patients who had not received additional antibiotic therapy. Their rate of TRM differed significantly from both early (log rank < .001) and late AB groups (log rank = .005, Figure 3A). Subgroup analysis revealed similar results in the Regensburg and New York cohorts (Table 3). Similarly, overall survival was lower in the early (51%, 121/236) than in the late (67%, 200/297; log rank < .001) and no AB groups (73%, 64/88; log rank = .001), respectively.

In the Regensburg cohort we observed a significantly higher rate of death from severe acute and/or chronic GVHD in the early (23%, 44/190) compared with the late AB (16%, 26/160) and the no AB groups (3%, 1/30; $P = .003$). The same association was seen in the New York cohort: Death from severe acute and/or chronic GVHD was observed in 26% (12/46) in the early, in 12% (16/137) in the late, and in 5% (3/58) in the no AB groups ($P = .01$).

Table 3
TRM in Relation to the Beginning of Antibiotic Treatment for Neutropenic Fever at the 2 Centers

	Total Cohort (N = 621)	Regensburg (n = 380)	New York (n = 241)
No AB group	7% (6/88)	3% (1/30)	9% (5/58)
Early AB group	34% (79/236)	33% (62/190)	37% (17/46)
Late AB group	21% (62/297)	19% (31/160)	23% (31/137)

TRM was significantly increased in the early AB group compared with the late AB group. The lowest rate of TRM was found in patients without additional antibiotic therapy.

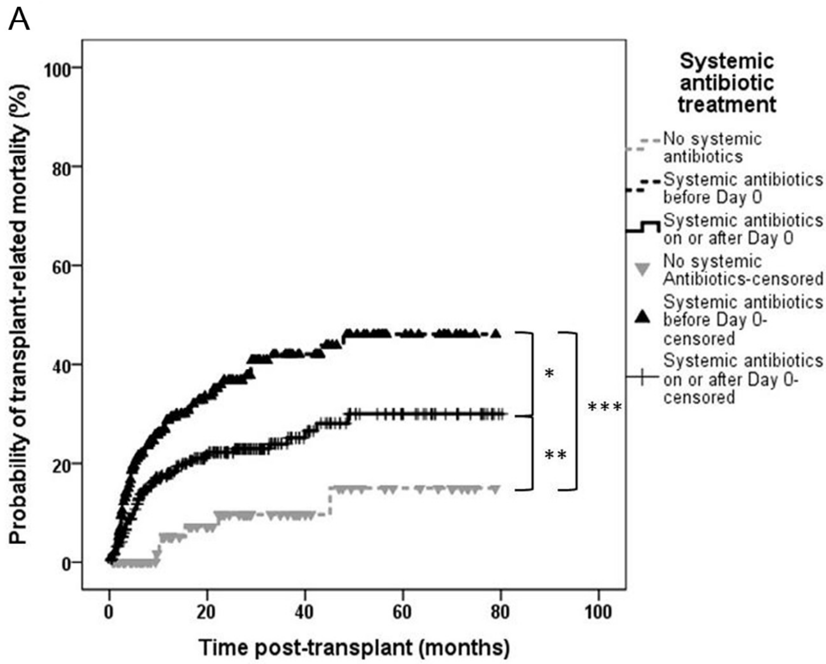
As previously reported in a smaller cohort of patients, we also observed an impact of antibiotic prophylaxis on TRM [16]. Patients who were given ciprofloxacin/metronidazole experienced a significantly higher rate of TRM (43/96, 44.8%) than patients administered rifaximin (19/94, 20.2%; $P < .001$).

It is possible that more heavily pretreated patients are both more susceptible to inferior long-term outcomes and to early peritransplant fevers that necessitate early antibiotic treatment. To address this potential confounder. we classified patients into acute leukemia versus nonacute leukemia groups and analyzed TRM rates. For nonacute leukemia patients, TRM was highest in the early AB group (37.0%, 37/100) followed by the late (25.1%, 42/167) and no AB groups (6.9%, 4/58; $P < .001$; Figure 3B). This held also true for acute leukemia patients independently (30.9%, 42/136; 15.4%, 20/130; and 6.7%, 2/30, respectively; $P = .002$; Figure 3C). In a multivariate analysis we identified early start of antibiotic treatment of neutropenic fever ($P < .001$), high patient age ($P = .004$), low Karnofsky performance status ($P = .03$), and female donor–male recipient sex combination ($P = .02$) as independent risk factors associated with TRM after ASCT (Table 4). The independent effect of early antibiotic treatment on GVHD-associated TRM ($P = .004$) could be demonstrated in a competing risk analysis, whereas infection-related TRM and relapse were not associated with start of systemic antibiotic treatment (not significant).

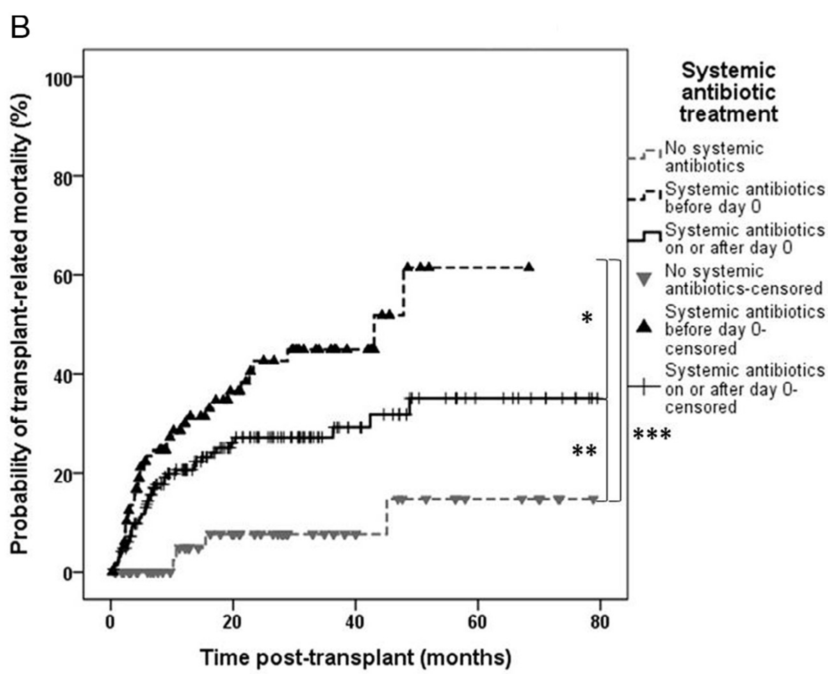
Table 4
Multivariate Risk Factor Analysis for TRM after ASCT: Early Initiation of Antibiotic Treatment, Patient Age > 40 Years, Karnofsky Performance Status < 90%, and Donor Female–Male Recipient Sex Combination Were Significantly Associated with Increased Risk of TRM

Risk factor	*P*	HR	95% CI for HR
Beginning of antibiotic treatment Early AB group	≤.001	2.0	1.42-2.83
Type of underlying disease Acute leukemia	.85	.97	.68-1.38
Patient age >40 yr	.004	2.15	1.27-3.64
Karnofsky performance status <90%	.03	1.47	1.05-2.06
Donor type Matched unrelated	.06	1.53	.99-2.36
Time from first diagnosis to ASCT <12 mo	.22	1.27	.87-1.85
Stage of underlying disease Advanced	.09	1.22	.97-1.54
Donor–recipient sex combination Donor female–recipient male	.02	1.56	1.06-2.29

Numbers for high-risk groups are indicated for categorical variables. Significance level <.05.



*p = 0.001; **p = 0.005; ***p = <0.001



*p = 0.016; **p = 0.005; ***p < 0.001

Figure 3. (A) Kaplan-Meier curves of TRM in relation to initiation of antibiotic treatment. Patients receiving antibiotic therapy before day 0 showed the highest TRM after ASCT. (B) Kaplan-Meier curves of TRM in relation to initiation of antibiotic treatment in the nonacute leukemia group. For patients without acute leukemia, TRM was highest in the early AB group followed by the late and no AB groups. (C) Kaplan-Meier curves of TRM in relation to initiation of antibiotic treatment in the acute leukemia group. TRM was highest in the early AB group compared with the late and no AB groups in patients with acute leukemia.

DISCUSSION

Our study reveals a novel association between the timing of antibiotic treatment, microbiota disruption, and outcome after ASCT. Risk factors for early antibiotic exposure included stage of underlying disease, donor type, and time interval from diagnosis to ASCT. The association between unrelated-donor ASCT and early antibiotic treatment may be due to an indirect association with ATG serotherapy, which is commonly given in the last days before an unrelated donor ASCT and often produces fever. This fever can trigger empiric antibiotic treatment because of the inability to differentiate between a real infection and a side effect of ATG in

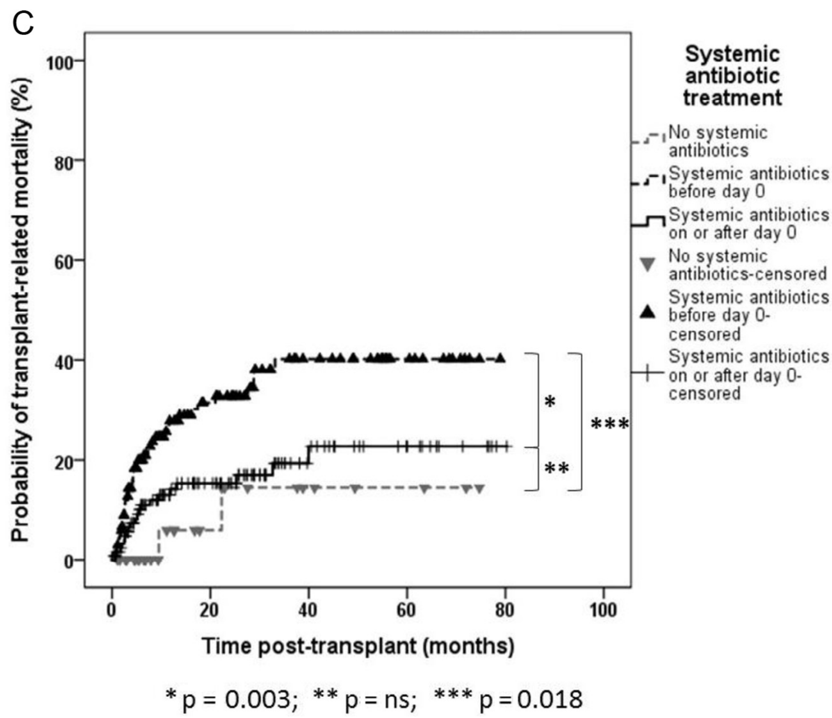


Figure 3. (*continued*).

high-risk neutropenic patients. In the Regensburg cohort, ATG was routinely administered before unrelated-donor transplantation, which might explain the higher proportion of patients in the early AB group compared with the New York cohort. Similarly, the correlation between early antibiotic exposure and short interval from diagnosis to ASCT might reflect the higher proportion of acute leukemia patients in the short-interval group who are more prone to neutropenic fever requiring early antibiotic treatment.

In this large study, early antibiotic treatment was associated with significantly altered microbiota composition. This lends further support to an association between early antibiotic-induced shifts in gut microbiota composition and long-term outcome after ASCT. As reported recently, urinary levels of 3-IS correlate with the abundance of *Clostridiales* in fecal specimens [6], and this effect on abundance of *Clostridiales* was confirmed in the New York group using a different approach of clostridial quantification. These associations were confirmed by *Clostridium* cluster XIVa species PCR in a small subgroup of patients.

Early antibiotic treatment was also correlated with a higher TRM and worse overall survival compared with patients treated with late or no additional broad-spectrum antibiotics. TRM rates in relation to initiation of antibiotic treatment were similar in the Regensburg and New York cohort, despite the use of 2 different antibiotic prophylaxis regimens. Furthermore, subgroup analysis showed that the correlation between early start of antibiotics and high TRM held true in both acute and nonacute leukemia subgroups. Although these data need to be confirmed by a prospective study, competing risk analysis identified the independent interaction of early antibiotic treatment and GVHD-associated TRM, whereas early exposure to antibiotics had no impact on infectious TRM. This analysis suggests that early exposure to antibiotics is not simply a surrogate parameter observed in more vulnerable patients who are at risk of early death from infectious complications. Rather, our data support the hypothesis that early antibiotic treatment induces microbiota disruptions and is itself a risk factor for worse outcome.

The lower TRM observed in patients treated with rifaximin compared with patients treated with ciprofloxacin/metronidazole is concordant with published observations of high antimicrobial activity of metronidazole against *Clostridium* cluster XIVa species [17] and of the major shifts in intestinal microbiota composition induced by ciprofloxacin [18]. Rifaximin exerts little effect on overall gut microbiota diversity and has in fact been reported to reduce expression of bacterial virulence factors, inhibit bacterial attachment to the intestinal mucosa, and decrease production of proinflammatory cytokines in the intestinal mucosa [19].

Our observation extends data reported on suppression of commensal *Blautia* species by antibiotic treatment in ASCT patients [9] and supports clinical and experimental data indicating that antibiotics with activity against anaerobic *Clostridiales* increase GVHD-related mortality [20]. Moreover, recent findings revealed that treatment of neutropenic fever with piperacillin-tazobactam or imipenem-cilastatin was associated with aggravated gut microbial perturbation and a significantly higher GVHD-related mortality compared with the use of aztreonam or cefepime, both antibiotics with reduced activity against anaerobic commensal bacteria in the gut [21]. Data from the present study add the important observation that not only does the type of antibiotic drug matter, but in particular the timing of antibiotic initiation has a crucial impact on microbiota composition and outcome after ASCT. We hypothesize that an early loss of protective bacteria in the gut because of early administration of systemic antibiotics affecting clostridia may result in a disturbed environment before the time of first donor T cell contact with recipient gastrointestinal tissues. The configuration of the intestinal microbiota in the immediate peritransplant period may have a more profound impact than a later microbiota disruption.

Several mechanisms have been identified regarding potential protective effects of commensal bacteria and may be relevant for our observation: In patients with early antibiotic treatment, the increased incidence of death from both chronic and acute GVHD indicates that long-term development of intestinal tolerance is impaired. Antibiotic treatment has been shown to suppress production of Reg3α in Paneth cells, which prevents colonization by pathogens [22,23]; similarly, metabolites of commensal *Clostridiales* mediate colonization resistance and maintain a microbiota supporting integrity of the intestinal epithelial barrier. Furthermore, commensal *Clostridiales* have been identified as important producers of short-chain fatty acids such as butyrate, which maintain intestinal regulatory T cells [24-26]. In addition, short-chain fatty acids can induce IL-22 responses in innate lymphoid cells, which exert anti-inflammatory and protective effects on intestinal epithelial cells [27]. Loss of these mechanisms may explain some of the long-term effects on GVHD-related TRM in our study. The time-dependent effects further indicate that a balanced gastrointestinal microbiota may be particularly critical in the early period after ASCT as donor lymphocytes migrate to peripheral target organs. The importance of a balanced intestinal microbiota for immunoregulation and the detrimental effects of antibiotics during critical periods is increasingly recognized for a variety of other human diseases: most convincingly, antibiotic exposure in early neonatal life has been shown to favor development of autoimmune diseases [28,29], and more recently even a negative effect of antibiotics on the efficacy of cytotoxic drugs such as cyclophosphamide or doxorubicin has been reported, as the microbiota can stimulate effector cells of the immune system that mediate optimal activity of cytotoxic treatment [30-32].

If confirmed in multicenter analyses, our observations have several implications. Because commensal *Clostridiales* are gram-positive anaerobic bacteria, the use of *Clostridiales*-sparing β-lactam antibiotics that target exclusively or mostly gram-negative bacteria for antibiotic therapy could first be considered. Depending on the clinical response of the patients, shorter durations of antibiotics seems feasible and might also contribute to more restricted use of antibiotics in low-risk patients. Second, prospective trials of antibiotic prophylaxis should be initiated to investigate different strategies or even no gut decontamination aiming at the protection of gut microbiota. Third, strategies to overcome the impact of antibiotics may be developed either based on pro- or on prebiotics applied in parallel to antibiotics. Alternatively, rapid restoration of microbiota diversity early after cessation of antibiotic treatment by either fecal microbiota transplantation or artificially composed microbiota seems a further feasible option. Beyond ASCT, our observations warrant a more critical use of antibiotics in general, not only to prevent the development of antibiotic resistance but also due to the observed indirect effects that commensal microbiota may have on long-term health in general.

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Authorship contribution: D.W. and E.H. were involved in conception and design of the study. D.Wo. and J.H. were responsible for collection of specimens. R.R.J., J.U.P., and Y.T. were responsible for data assessment and data analysis of the New York cohort. P.J.O. and K.D. performed measurements of 3-IS levels. A.H., J.K., and A.G. conducted quantitative reverse transcriptase PCR of *Clostridium* cluster XIVa species. M.W. contributed to statistical data analysis. D.W. and E.H. collected and analyzed clinical data and wrote the manuscript. W.H., M.R.M.B., and E.P. were involved in interpretation and discussion of study results according to the current literature. All authors read, revised, and approved the final draft.

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